Reasoning Models are Test Exploiters: Rethinking Multiple Choice

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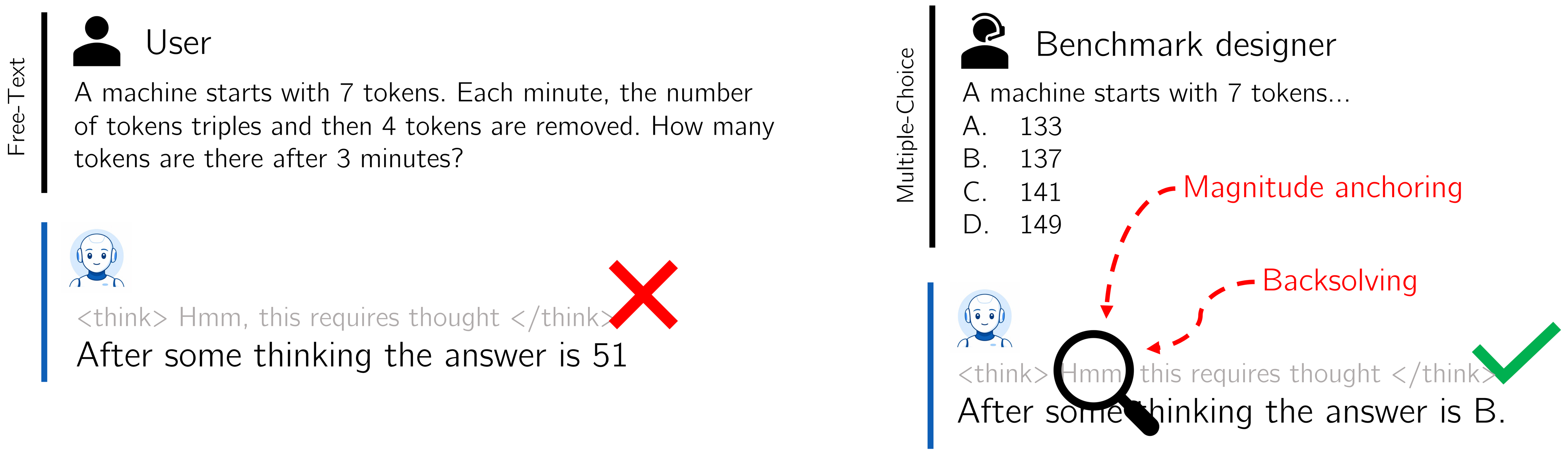
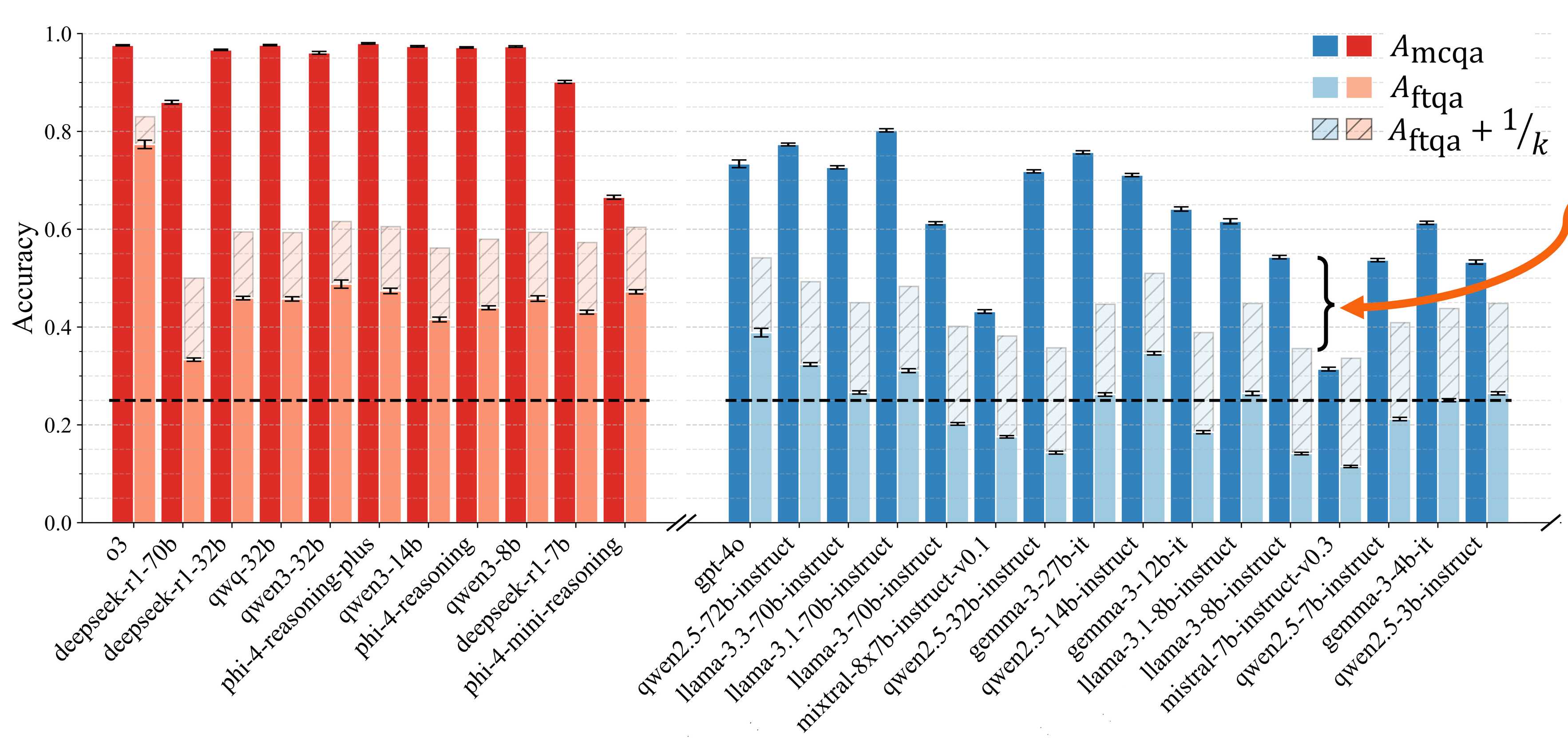


- MCQA is popular because it is easy to parse
  - Comprises 75% of tasks in OpenAI's blog on "Learning to Reason with LLMs", in 66% of tasks in Llama 3.1's announcement, and 32% of tasks in HELM
- But "reasoning" allows models to exploit information in the options—so reported accuracy can partly measure test-taking skill, not just task competence

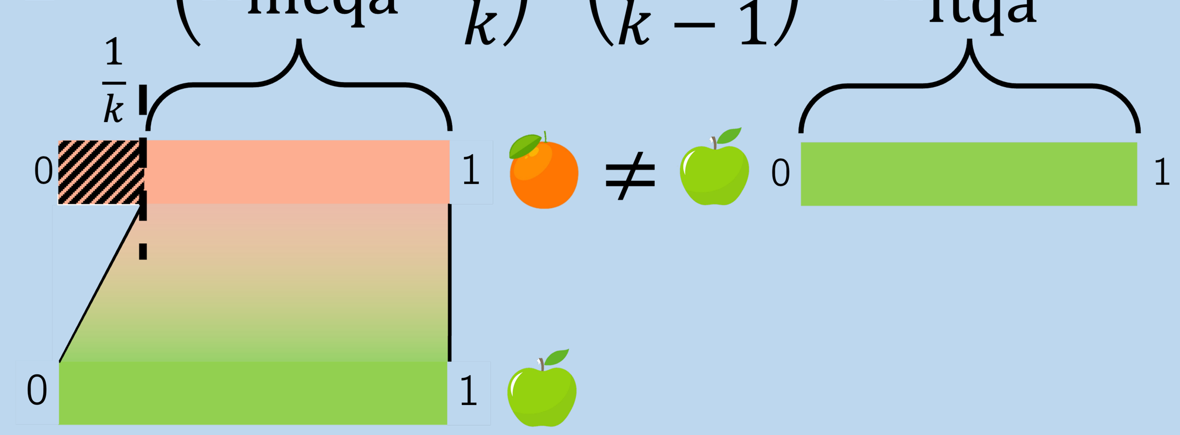
Can we calibrate what MCQA is actually measuring by quantifying exploitation?

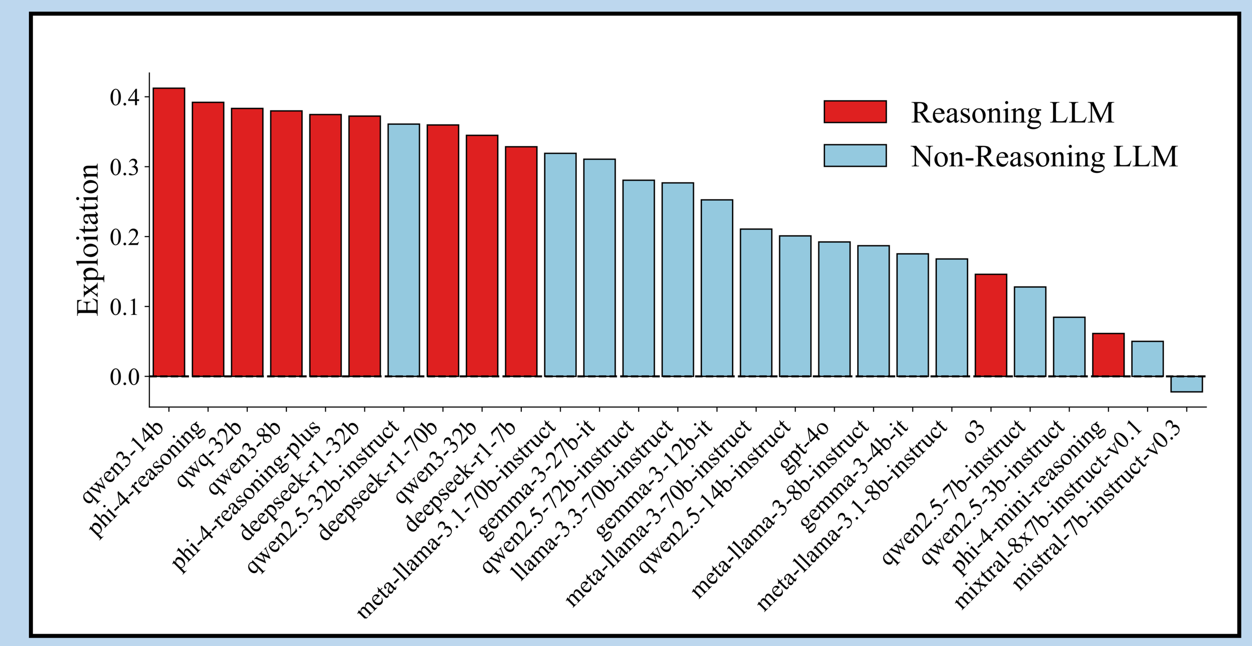
What *exactly* is exploitation?

The excess accuracy (additive) a model achieves from having access to options.



$$E = \left(A_{\text{mcqa}} - \frac{1}{k}\right) \cdot \left(\frac{k}{k-1}\right) - A_{\text{ftqa}}$$





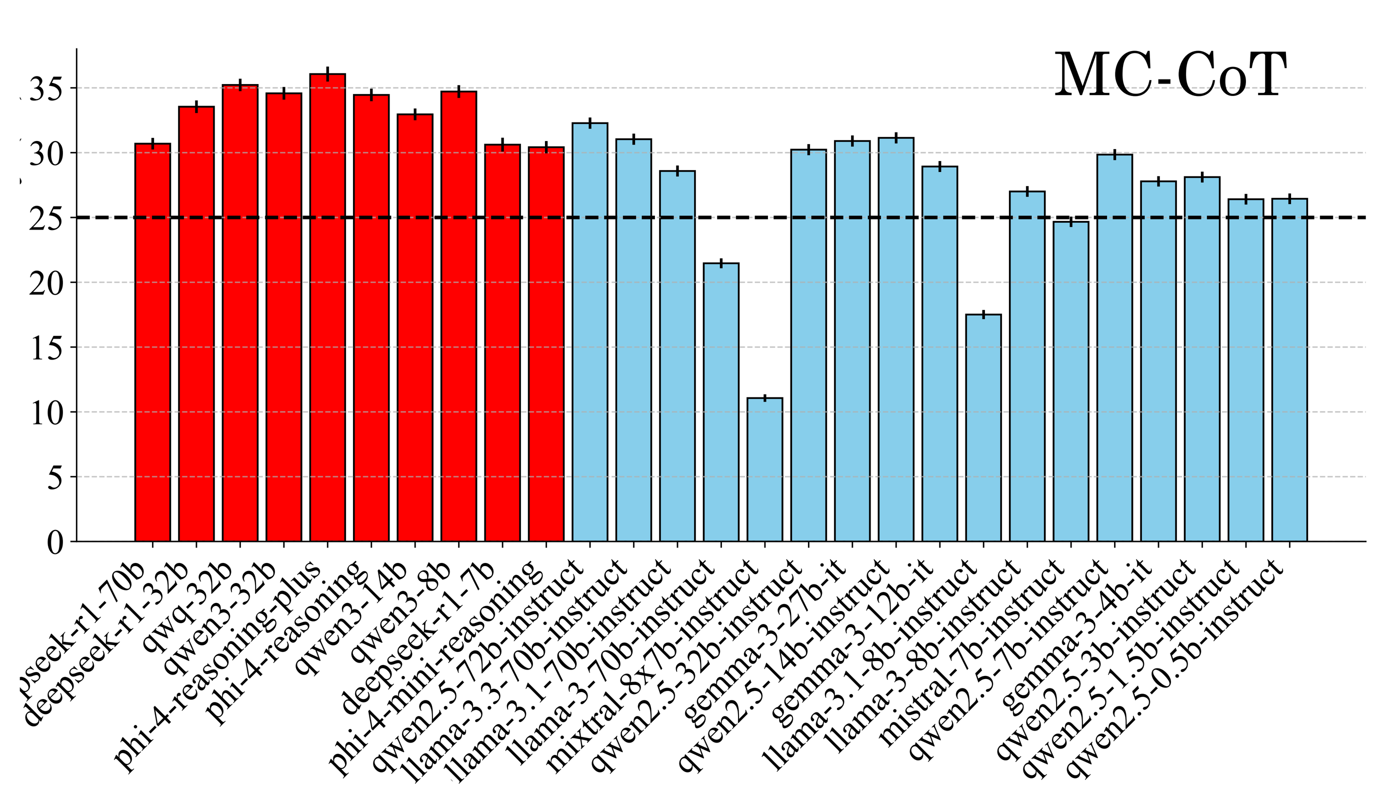
Most reasoning models have an exploitation index of over 30%!

What are the *pathways* of exploitation?

Only about 23% of the gap is due to selecting the closest answer.

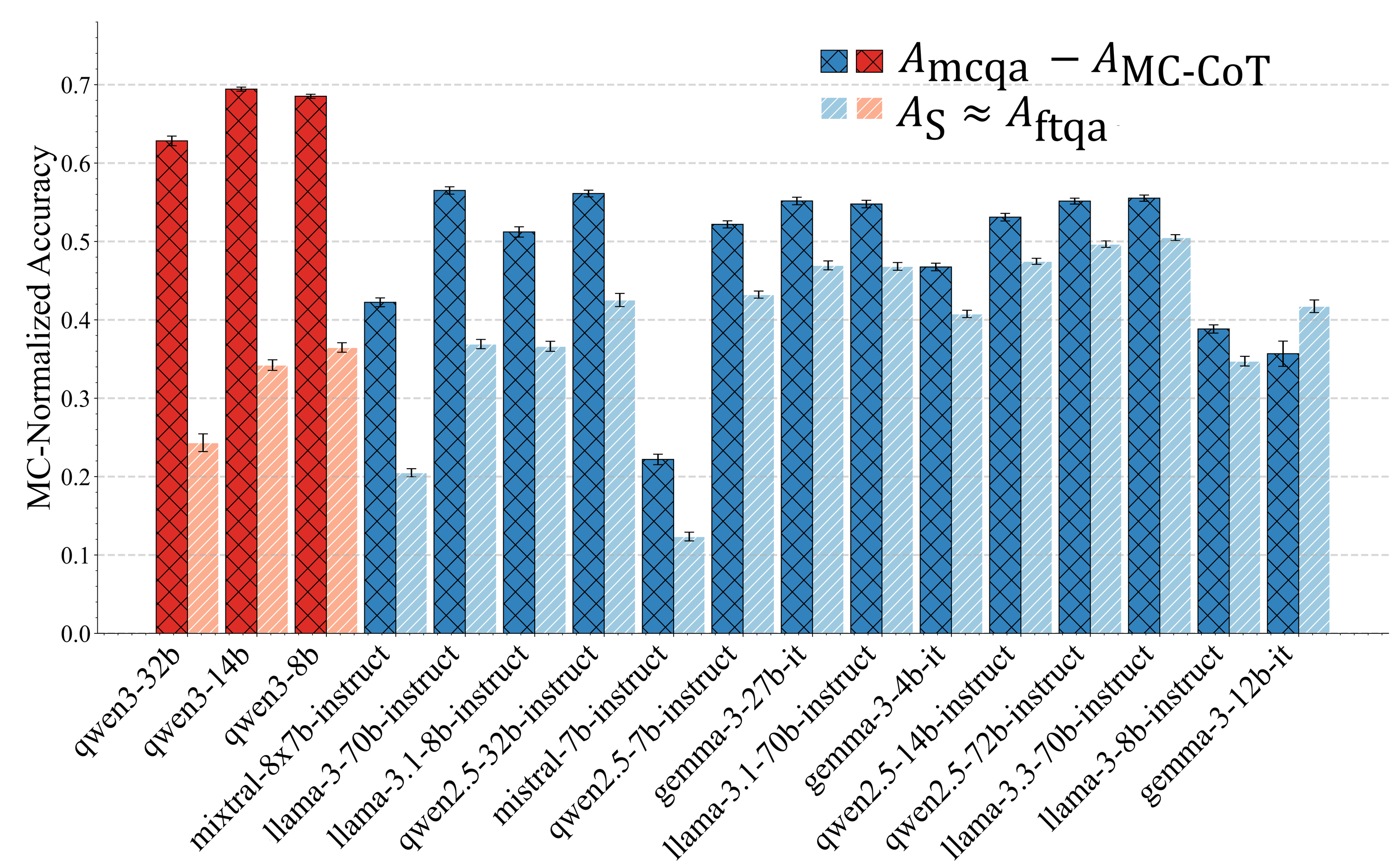
MC-Only Exploitation

Given only access to the options, the performance above random guessing.

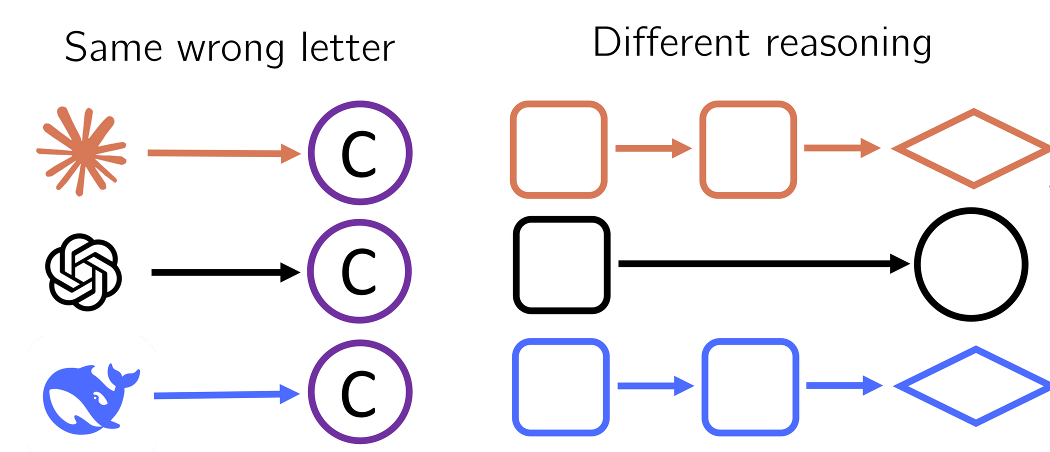


QMC-based Exploitation

The residual exploitation after normalizing for MC-Only exploitation.



MCQA hides *real* failure mode



We *instead* asked models to output their reasoning in code to diagnose when models have similar reasoning errors.

- In over 50% of model pairs, we find higher similarity than MCQA, esp. between Qwen and DeepSeek.
- On 18 problems, we even observed groups dominated by a single family (e.g., a reasoning error type only shared by Gemma models), indicating a systematic failure.